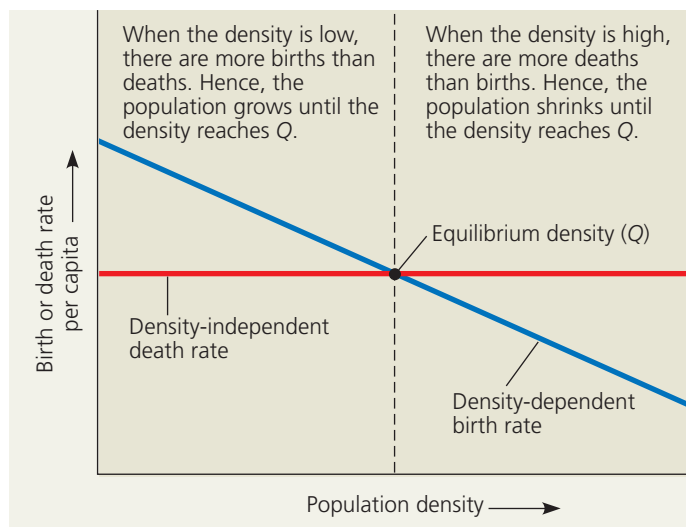


each other, then a population grows when the birth rate exceeds the death rate and declines when the death rate exceeds the birth rate.

A birth rate or death rate that does *not* change with population density is said to be **density independent**. For example, researchers found that the mortality of dune fescue grass (*Vulpia fasciculata*) is mainly due to physical factors that kill similar proportions of a local population, regardless of its density. Drought stress that arises when the roots of the grass are uncovered by shifting sands is a density-independent factor that can kill these plants. In contrast, a death rate that increases with population density or a birth rate that falls with rising density is said to be **density dependent**. Researchers found that reproduction by dune fescue declines as population density increases, in part because water or nutrients become more scarce. Thus, the key factors affecting birth rate in this population are density dependent, while death rate is largely determined by density-independent factors. For situations in which immigration and emigration offset each other, **Figure 53.15** shows how the combination of density-dependent reproduction and density-independent mortality can stop population growth (as in species such as dune fescue).

Variation in density-independent factors such as temperature and precipitation can cause dramatic changes in population size. For example, a drought or heat wave can cause a sharp increase in mortality rates, causing the abundance of a population to plummet. Note, however, that

▼ **Figure 53.15 Determining equilibrium for population density.** In this example, the birth rate changes with population density, while the death rate is constant; immigration and emigration rates are assumed to be equal. At the equilibrium density (Q), the birth and death rates are equal. As a result, the number of individuals added to the population equals the number removed from it, and the population stops changing in size.



DRAW IT Redraw this figure for the case where the birth and death rates are both density dependent, as occurs for many species.

a density-independent factor cannot consistently cause a population to decrease in size when it is large or increase in size when it is small—only a density-dependent factor can consistently cause such changes. With this in mind, a population is said to be *regulated* when one or more density-dependent factors cause its size to decrease when large (or increase when small).

➔ **Mastering Biology BioFlix® Animation: Density Dependence**

Mechanisms of Density-Dependent Population Regulation

Without some type of negative feedback between population density and the rates of birth and death, a population would never stop growing. But no population can increase in size indefinitely. Ultimately, at large population sizes, negative feedback is provided by density-dependent regulation, which halts population growth through mechanisms that reduce birth rates or increase death rates. For example, a study of kelp perch (*Brachyistius frenatus*) populations showed that the fish's death rate rose proportionally as its density increased (**Figure 53.16**). This occurred because at high densities, the fish ran out of safe spaces in the kelp where they could hide from predators. Several other mechanisms that can cause density-dependent regulation are described in **Figure 53.17**.

These various examples of population regulation by negative feedback show how increased densities cause population growth rates to decline by affecting reproduction, growth, and survival. Although negative feedback helps explain why populations stop growing, it does not address why some populations fluctuate dramatically while others remain relatively stable.

▼ **Figure 53.16 An example of density-dependent regulation.** At high population densities, the death rate of kelp perch increased because they ran out of spaces to hide from predators.

